MCKTNet: Multiscale Cross-Modal Knowledge Transfer Network for Semantic Segmentation of Remote Sensing Images

Jian Cui, Jiahang Liu, *Member, IEEE*, Yue Ni, Yuan Sun, and Mao Guo

***Abstract*— Multimodal data fusion can provide valuable and diverse information for remote sensing image segmentation. However, different modal data have different feature distributions, which causes some conflicts and redundancies in cross-modal feature fusion. In addition, existing multimodal fusion networks usually adopt a two-branch structure with a large number of parameters and high computational cost. To address these problems, we propose a multiscale cross-modal knowledge transfer network (MCKTNet) for remote sensing image segmentation. First, we use a cross-modal migration learning method that combines channel discrete loss and spatial discrete loss to facilitate cross-modal migration of geometric and semantic features and reduces redundancy by minimizing the differences in feature distribution. Then, we use the polarized cross-self-attention mechanism to establish long-range correlations between different modal features across spatial and channel dimensions, and only a small number of parameters need to be added to achieve complementary cross-modal feature fusion. Finally, to accurately capture object edges, we propose a multiscale edge perception module to optimize the edge details in the prediction results at the pixel level. Extensive experiments demonstrate that the proposed method is effective, robust, and generalizable, and achieves state-of-the-art performance in multiple remote sensing image semantic segmentation tasks with only 13.51 million parameters. The code will be available at https://github.com/NUAALISILab/MCKTNet.**

***Index Terms*— Multimodal, remote sensing, semantic segmentation, transfer learning.**

I. Introduction

SEMANTIC segmentation of remote sensing images, also known as land-cover classification, aims at locating objects at the pixel level and classifies each pixel in a remote sensing image into a specified category [1]. It plays an important role in many remote sensing applications, such as urban planning [2], [3], smart cities [4], [5], and geographic information system construction [6]. Considering ease of use, most methods use unimodal data for segmentation. Due to the singularity of information, it is difficult to achieve satisfactory results in complex scenes using only unimodal data. Multimodal data can provide richer information from different perspectives and thus has obvious potential for complex scenes and high-precision segmentation, which is a new way to solve remote sensing image segmentation.

Fusing multimodal data to achieve semantic segmentation of remote sensing images can make full use of the complementary features of different data types to obtain more comprehensive and rich semantic information, thereby improving the accuracy and reliability of semantic segmentation [7]. Taking Fig. 1 as an example, due to factors such as lighting, seasons, and crop varieties, there may be significant variations in the feature distributions among farmland areas in multispectral images, which can sometimes closely resemble the features of ponds. Solely relying on multispectral images is insufficient for effective differentiation. Extracting effective features from the normalized difference vegetation index (NDVI) and integrating them with the rich texture features present in multispectral images can enhance the performance of vegetation detection models in complex scenarios. Therefore, effectively extracting complementary information from different modalities to obtain richer features than those derived from a single data source has become a new technical challenge in the field of remote sensing data processing.

An efficient feature representation serves to enhance the discriminative capacity of category information [8]. Constructing the feature space of a multisource segmentation model by fully exploring the complementary and discriminative features of different data sources is key to achieving high-precision segmentation results of multimodal remote sensing images.

To extract complementary features among multimodal data, existing models have explored incorporating attention mechanisms [9] or gate mechanisms [10], [11] during feature fusion for information filtering. However, these methods focused on the correlation among diverse modal features, while the impact of their inherent data distribution disparities on the fusion effectiveness is overlooked. The research shows that multimodal data generated by different sensors exhibit vastly distinct characteristics, such as cross-modal heterogeneous statistical properties and noise levels, and the effectiveness of feature fusion decreases with the increase of feature distribution difference between different modal data [33]. Unlike

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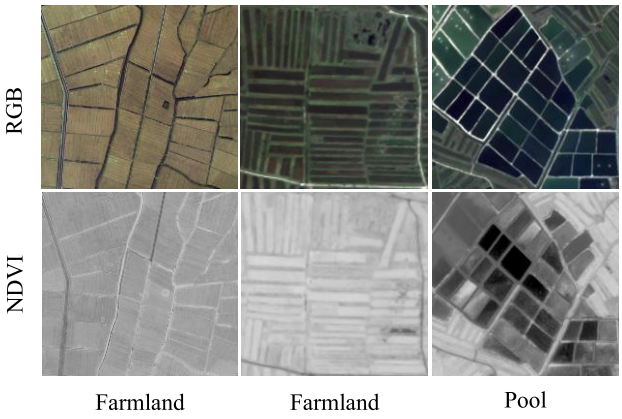


Fig. 1. Different modal data in the same scene have different distributions of features.

the semantic segmentation tasks for ordinary scene images (such as cars and people), remote-sensing image targets (such as water bodies, vegetation, and so on) often lack consistent boundary features. In addition, the processes of downsampling and multimodal feature fusion can both lead to the loss of spatial detail features in low-level features, resulting in inaccurate segmentation of the edge parts of the objects. Due to the necessity of employing a dual-branch architecture for multimodal feature extraction in multimodal fusion models, these models tend to possess a large number of parameters and complex computations, which hinders their deployment and practical application.

To address the above issues, a multiscale cross-modal knowledge transfer network (MCKTNet) is proposed for the semantic segmentation of remote sensing images. We design a cross-modal transfer learning method to transfer semantic and geometric features between different modal data to reduce the differences in feature distribution among different modal data. The cross-attention mechanism is used to achieve the complementary fusion of cross-modal features. To achieve more accurate object edges, we design a multiscale edge-aware optimization module to guide the operation to directly optimize the edge details from the prediction result at the pixel level. MCKTNet employs a lightweight network framework that harnesses knowledge transfer to alleviate the data distribution discrepancies between diverse modalities. It minimizes the redundancy between these modalities by performing bidirectional migration of semantic and geometric information. Distinct from traditional cross-modal knowledge distillation (KD) methods based on teacher-student networks, this approach directly enhances the representation of the backbone features through simultaneous bidirectional knowledge transfer, without the need to pre-train a teacher model for the network or components. The main contributions of this work can be summarized as follows.

1) We have designed a multimodal fusion method based on cross-modal knowledge transfer. This method employs distinctively the channel-wise discrete loss module and the spatial discrete loss module to facilitate the migration of geometric and semantic features between different modal data, thereby minimizing the redundancy among various modalities.
2) We presented an efficient multimodal cross-self-attention mechanism to facilitate sufficient information exchange between interactive vectors from diverse modal paths.
3) A multiscale edge optimization module is proposed, which directly extracts edge information from pixel-level segmentation predictions. This module optimizes the prediction outcomes without incurring additional parameters or computations, thereby mitigating the information loss of low-dimensional features and enhancing edge semantic features.

Experimental results on different datasets validate the excellent feature fusion capability of our proposed MCKTNet for different models. Compared with other state-of-the-art methods, MCKTNet can efficiently integrate different prior knowledge and achieve the best accuracy in single-class segmentation tasks exemplified by buildings, vegetation, and water bodies.

The remainder of this article is organized as follows. Section II first reviews the related works on remote sensing segmentation, modality fusion, and KD methods. After that, Section III presents the structure of the proposed MCKTNet, whereas Section IV provides details on the extensive experiments conducted. Finally, the conclusion is given in Section V.

II. Related Works

A. Remote Sensing Image Semantic Segmentation

With the outstanding results of convolutional neural networks for computer vision tasks, more and more researchers have started to focus on deep learning-based remote sensing image semantic segmentation. Xu et al. [12] propose a lightweight dual attention module to obtain image-level contextual information, and then the feature enhancement feature pyramid module is designed to exploit the local-level and image-level contextual information in parallel structure. To address the problem of intra-class differences, Zhang et al. [13] aggregated local-level contextual information in a high-resolution branch and harvested image-level contextual information in a low-resolution branch based on HRNet. Dong et al. [14] proposed DenseUNet for small object semantic segmentation in urban remote sensing images, where DenseUNet connects the CNN features through cascade operations and fuses the detail features in the shallow layers, as well as the abstract semantic features in the deep layers. Ma et al. [15] proposed a foreground activation-driven small object semantic segmentation framework for HRS imagery. Li et al. [16] proposed a multiattention network to extract context dependencies through multiple attention modules to leverage multiscale information. In addition, to address the accurate distinction of edge areas between different classes, Wang et al. [48] proposed a key feature repairing network (KFRNet) based on self-supervised learning (SSL), which calculates patch similarities to identify and mask prominent features for repair, and incorporates an image comparison branch to obtain category features by contrasting positive and negative samples, thereby improving segmentation accuracy. Li et al. [49] proposed a novel boundary-enhanced dual-stream network (BEDSN), which integrates an edge detection branch stream (EDBS) with a composite loss function and a

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semantic segmentation branch stream (SSBS) through a highly coupled encoder and feature extractor, and enhances SSBS with multiscale feature extraction and hybrid atrous convolution modules. Although these methods have achieved good performance, they are based on a single source input which is not stable in different cases and limits further improvement.

B. Modality Fusion in Remote Sensing Segmentation

Recent studies have shown that multiple sources [such as HSI, MSI, near-infrared (NIR), and LiDAR] features can effectively promote the robustness of scene parsing [10], [11], [17]. In particular, the encoder–decoder architecture has been shown very effective in fusing multimodal data. According to where the fusion takes place, these encoder–decoder methods can be classified into three approaches, namely data-level fusion, middle fusion, and late fusion. Data-level fusion approaches combine spectral information such as red, green, blue, and NIR spectrum with structural information such as the digital surface model (DSM) as the input to a network. For example, ResUNet-a uses the RGB and DSM images by directly concatenating the images in the data level before feeding the concatenated data into encoders [18].

In contrast, in the middle fusion methods spectral and structural information are sent to separate identical encoders for each modality, and the lateral features of two encoders are merged in the cross-modal using concatenation operation or element-wise summation at different scales. For example, Zhou et al. [19] proposed a gate mechanism to filter out redundant information from multimodal data and used their similarities to extract complementary information for feature fusion. Finally, in late fusion methods [20], different data sources are sent to the two different encoder–decoder network paths, and predicted classes are fused in the final stage. In current advanced multimodal fusion methods, the emphasis is placed on the correlation among diverse modal features, while the impact of their inherent data distribution disparities on fusion effectiveness is overlooked. To reduce the distribution discrepancies in cross-modal data, Yang et al. [50] proposed a multimodal fusion method that integrates hierarchical feature aggregation and cross-modal feature alignment to address the challenges of feature representation and alignment in remote sensing image captioning (RSIC). However, this method only relies on cross-modal feature alignment loss to minimize the distribution difference between features, which ignores the inherent characteristics of different data types and leads to insufficient utilization of cross-modal data features.

C. Knowledge Distillation

KD provides a new concept for lightweight models with high segmentation performance. It extracts specific knowledge from large, bulky teacher networks and transfers it to lightweight, small student networks. Hinton et al. [21] were the first to propose a KD method to improve the segmentation performance of a small network by efficiently softening the target on the final prediction layer. Then, Romero et al. [22] built on this foundation by proposing knowledge transfer to intermediate features. Thereafter, an increasing number of KD schemes have been proposed, which can be roughly classified into three categories: relation-based [23], response-based [24], and feature-based [25]. Chen et al. [26] first proposed the concept of learning from the past, in which previous feature information is continuously added to the review during the distillation process to improve the correlation between features. Feng et al. [27] used the residual attention mechanism to capture the similarity between teachers and students in both pixel- and class-level dimensions. Yang et al. [28] used global semantic relations on the feature map to transfer structured knowledge from the teacher model to the student model. Zhang et al. [29] proposed an interclass distance distillation scheme, believing that the accuracy of interclass relationships was very important for segmentation performance.

These methods need to train a teacher network model first, and then train the student network to make the student network close to the teacher network to the greatest extent. They have high requirements on the performance of the teacher network, and can only learn the fixed model in a single direction, and the cross-modal features are not fully learned.

III. Methodology

A. Framework Overview

This article proposes a lightweight MCKTNet, which leverages cross-modal knowledge transfer and cross-self-attention mechanisms to achieve effective fusion of multimodal data.

The overall architecture of the proposed MCKTNet is shown in Fig. 2(a). It has a typical encoder–decoder structure. In the encoder, to adapt MobileNetV2 [30] pre-trained on the ImageNet dataset to semantic segmentation of remote sensing images, we remove the global average pooling and the last fully connected layers from the backbone and obtain five scales of outputs with various dimensions: $16 \times 128 \times 128$, $24 \times 64 \times 64$, $32 \times 32 \times 32$, $96 \times 16 \times 16$, and $320 \times 8 \times 8$ (number of channels $\times$ height $\times$ width). For these five outputs, the near infrared, red, green, blue (IRRGB) path and prior data path are represented by r_a and p_a, respectively, where $a = 1, 2, 3, 4, 5$. As shown in Fig. 3, cross-modal geometric semantic feature transfer (CGSF) exploits the complementarity between different modal data to exchange semantic and geometric information during testing without incurring additional computations. In addition, MCKTNet includes the boundary optimization algorithm in the decoder d_3, d_4, and d_5 to compensate for limitations of the lightweight backbone at the spatial level without incurring additional parameters and computations.

B. Cross-Modal Geometric Semantic Feature Transfer Module

CGSF module can effectively carry out feature exchange between different modal data, so that the data distribution of different modes is closer, and reduces the data redundancy problem in the process of cross-modal fusion. CGSF is divided into two different structures, the cross-modal geometric feature transfer (CGF) module and cross-modal semantic feature transfer (CSF) module, which are used for the fusion of texture features and semantic features, respectively.

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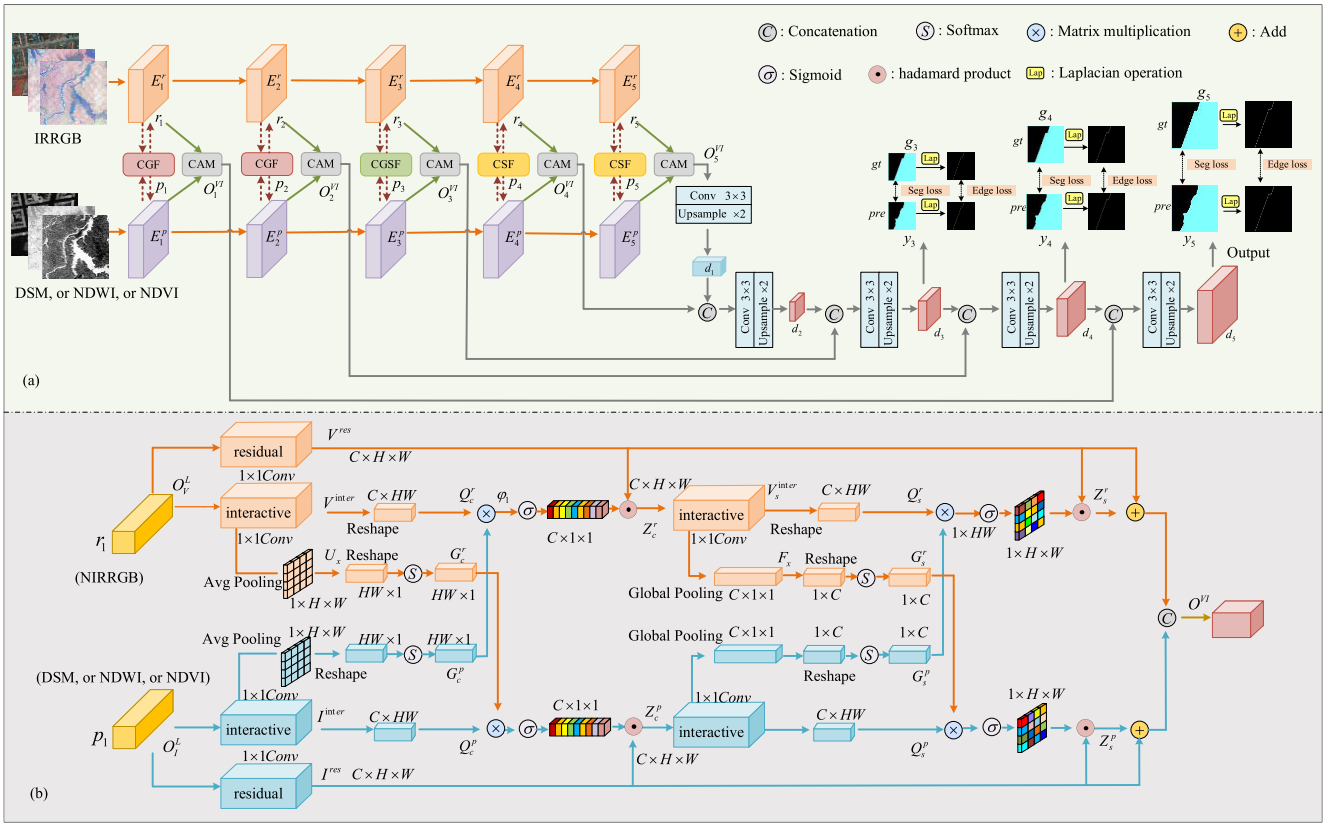


Fig. 2. (a) Overview of MCKTNet for multimodal semantic segmentation. The inputs are an IRRGB image and another modality data (e.g., DSM, NDWI, and NDVI). (b) Detailed architecture of the multimodal CAM.

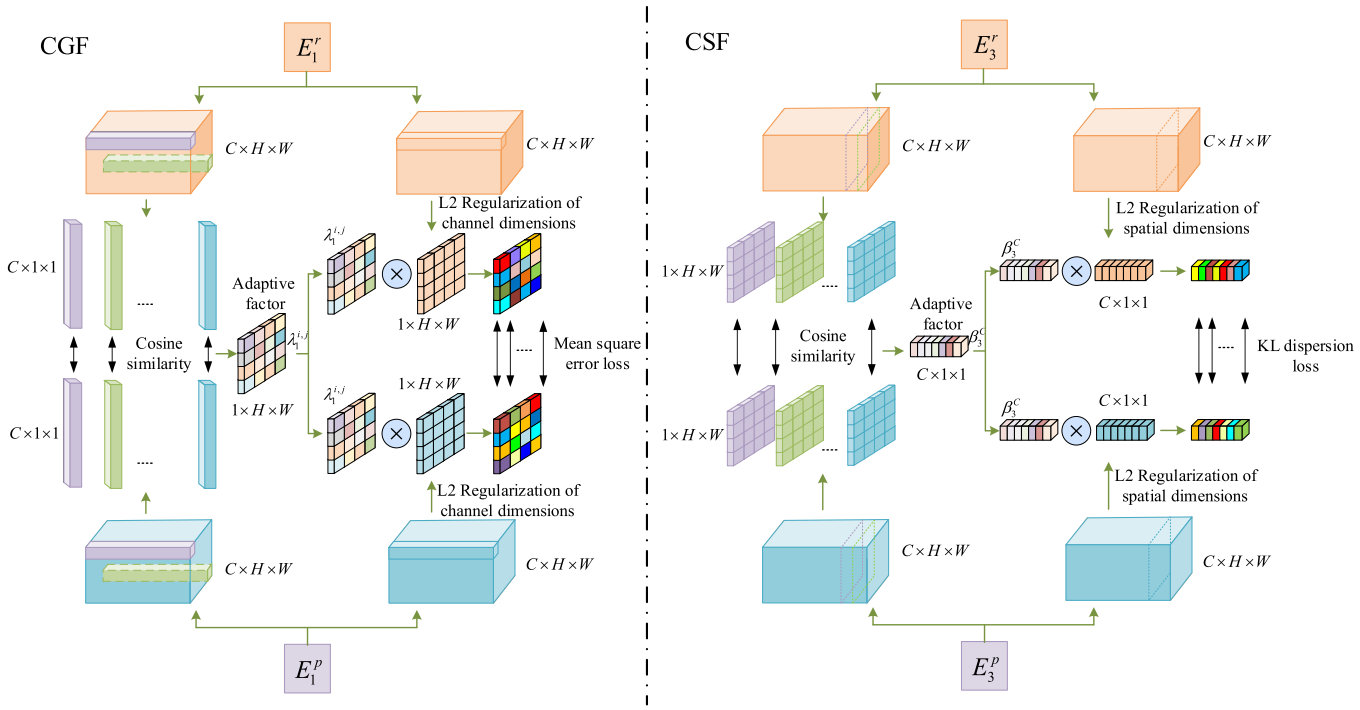


Fig. 3. Detailed architecture of the CGSF, including CGF and CSF.

CGF: In CNN, low-level features represent spatial and structural information. Accordingly, we propose CGF inspired by the development in [31] to leverage attentive transfer learning of low-level spatial features in the IRRGB and Prior data. According to the research in [32] and [33], the magnitude of feature differences among various modalities of data

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serves as the primary factor influencing the effectiveness of knowledge transfer. Specifically, a greater discrepancy in data features leads to a poorer outcome in knowledge transfer and directly learning cross-modal data distributions from vastly different modalities may introduce adverse effects in segmentation prediction. Consequently, we propose an adaptive feature selection scheme to ensure the efficacy of cross-modal knowledge transfer. More precisely, we devise an adaptive factor (AF) λ to modulate the influence of different modal paths. When significant discrepancies exist in the feature distributions of various modal paths, the extent of knowledge transfer is reduced.

We select the output $r_a(a = 1, 2, 3)$ from the IRRGB path and the output $p_a(a = 1, 2, 3)$ from the prior data path as shallow features to perform CGF. Taking r_1 and p_1 as an example, for any corresponding points r_1 and p_1 on the plane $H \times W$ (where H, W represent the height and width of the feature map), we calculate the distance R between the vectors $r_1^{i,j}$ and $p_1^{i,j}$, which are composed of all points located at the positions of points $r_1(i, j)$ and $p_1(i, j)$ across all channels C. Theoretically, R can be any metric for measuring the distance between vectors, such as Euclidean distance or cosine distance. Given that the dimensionality of feature vectors can be extremely high, to alleviate the curse of dimensionality, we choose cosine similarity R. The cosine similarity between vectors $r_1^{i,j}$ and $p_1^{i,j}$ is computed as an indicator to evaluate the magnitude of feature differences between different modalities. Subsequently, the value $R$is transformed into the range [0, 1] to obtain the AF $\lambda_1^{i,j}$

$$R\left(r_1^{i,j}, p_1^{i,j}\right) = \frac{\left(r_1^{i,j}\right)^{\mathrm{T}} p_1^{i,j}}{\|r_1^{i,j}\|_2 \|p_1^{i,j}\|_2} \tag{1}$$

$$\lambda_1^{i,j} = \frac{R\left(r_1^{i,j}, p_1^{i,j}\right) + 1}{2} \tag{2}$$

where (i, j) represents a value in (H, W). Cosine similarity serves a gradual fusion function, preventing rapid and significant changes in features when feature differences are too large, thereby maintaining the utilization of the modality's inherent features.

We apply the $L2$ norm before transfer learning for the features to have a similar range and simplify learning

$$r_1^{norm} = L2norm(r_1) \tag{3}$$

$$p_1^{norm} = L2norm(p_1). \tag{4}$$

The $L2$ normalization is applied along the channel dimension, transforming the feature dimensionality from $C \times H \times W$ to $1 \times H \times W$, to preserve the geometric information of the spatial structure. As this module necessitates the assignment of an AF to each element at every spatial location on the feature map during cross-modal knowledge transfer, we employ mean squared error (mse) loss within this module to quantify the differences in feature distributions across different modalities

$$l_1^{\mathrm{mse}} = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \lambda_1^{i,j} \times \left(r_{1_{i,j}}^{norm} - p_{1_{i,j}}^{norm}\right)^2 \tag{5}$$

where (i, j) represents a value in (H, W). Since $r_a(a = 1, 2, 3)$ and $p_a(a = 1, 2, 3)$ use the same structure in the CGF module, the loss function for the CGF can be expressed as follows:

$$l_{CGF} = \sum_{a=1}^{3} l_a^{\mathrm{mse}}, \quad a = 1, 2, 3. \tag{6}$$

CSF: The CSF emphasizes the transfer of high-level semantic information. We choose the output $r_a(a = 3, 4, 5)$ of the IRRGB path and the prior data path $p_a(a = 3, 4, 5)$ as deep features to carry out cross-modal semantic information transfer. Take r_3 and p_3 as an example, calculate the cosine similarity $R(F_{r_3}^i, F_{p_3}^i)$ of the feature graph $F_{r_3}^i$ and $F_{p_3}^i$ of r_3 and p_3 on channel C_i, where $i \in C$, and then transform the value of $R(F_{r_3}^i, F_{p_3}^i)$ into the range [0, 1] to obtain the AF β_3^i

$$\beta_3^i = \left(\frac{\left(F_{r_3}^i\right)^{\mathrm{T}} F_{p_3}^i}{\|F_{r_3}^i\|_2 \|F_{p_3}^i\|_2} + 1\right)/2 \tag{7}$$

where i denotes any arbitrary channel within the channel set C. By computing the cosine similarity between the feature maps of all corresponding channels on r_3 and p_3, we obtain a channel similarity vector $C \times 1 \times 1$ with a dimension of β_3^C.

Similarly, $L2$ normalization is applied to r_3 and p_3 in the spatial dimension, resulting in r_3^{norm} and p_3^{norm}, respectively. The $L2$ normalization along the spatial dimension can transform the feature dimensionality from $C \times H \times W$ to $C \times 1 \times 1$, thereby preserving the global semantic information of each channel. Subsequently, the KL divergence loss is utilized to calculate the difference in feature distributions between r_3^{norm} and p_3^{norm}, and the AF β_3^i is assigned to each channel

$$l_{r_3,p_3}^{\mathrm{CSF}} = \beta_3^C \times D_{\mathrm{KL}}\left(r_3^{norm} || p_3^{norm}\right). \tag{8}$$

Since the KL divergence calculation is not symmetric, we also need to switch the positions of r and p to compute the KL divergence loss between p_3^{norm} and r_3^{norm}

$$l_{p_3,r_3}^{\mathrm{CSF}} = \beta_3^C \times D_{\mathrm{KL}}\left(p_3^{norm} || r_3^{norm}\right). \tag{9}$$

Since $r_a(a = 3, 4, 5)$ and $p_a(a = 3, 4, 5)$ use the same structure in the CSF module, the loss function for the CSF can be expressed as follows:

$$l_{r,p}^{\mathrm{CSF}} = \sum_{a=3}^{5} l_{r_a,p_a}^{\mathrm{CSF}}, \quad a = 3, 4, 5 \tag{10}$$

$$l_{p,r}^{\mathrm{CSF}} = \sum_{a=3}^{5} l_{p_a,r_a}^{\mathrm{CSF}}, \quad a = 3, 4, 5 \tag{11}$$

$$l_{\mathrm{CSF}} = l_{r,p}^{\mathrm{CSF}} + l_{p,r}^{\mathrm{CSF}}. \tag{12}$$

C. MultiModal CAM

We designed a multimodal feature correlation degree fusion module to merge the features from the two modalities and enhance cross-modal information interaction. As shown in Fig. 2(b), we introduce an efficient multimodal cross-self-attention mechanism to facilitate sufficient information exchange between interactive vectors from diverse modal paths. To address the issues of high parameter quantity and complex computation in the self-attention algorithm, our multimodal cross-self-attention mechanism employs a lightweight polarized self-attention mechanism [34], which establishes correlation features between different modal data from both spatial and channel dimensions.

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For brevity, we take the IRRGB multispectral image path for illustration. We first utilize two 1×1 convolutions to extract features from input features with size $O_V^L \in \mathbb{R}^{C\times H\times W}$, resulting in two feature vectors referred to as the residual vector V^{res} and the interaction vector V^{inter}.

Average pooling is applied to the interactive vector V^{inter} across the channel dimension to generate the feature vector U_x. The dimension of U_x is altered using the projection function $\theta(\cdot) : \mathbb{R}^{1\times H\times W} \rightarrow \mathbb{R}^{HW\times 1}$. Subsequently, the vector G_c^r is obtained by employing the Softmax activation function, thereby enhancing the dynamic range of feature activation. The calculation formula is provided below

$$G_c^r = \text{Soft}\max(\theta(U_x)). \tag{13}$$

Then, the projective function $\mu(\cdot) : \mathbb{R}^{C\times H\times W} \rightarrow \mathbb{R}^{C\times HW}$ is used to change the dimension of the interactive vector V^{inter}, to get the vector $Q_c^r = \mu(V^{\text{inter}})$. The prior path and the multispectral path adopt the same network structure, and the vectors Q_c^p and G_c^P of the prior path are obtained using the same method.

The features G_c^P and Q_c^r are subjected to similarity calculation to obtain a similarity matrix $\varphi_1 = Q_c^r \otimes G_c^P$, where $\otimes$ is matrix multiplication. Then, the sigmoid function is applied to normalize the vector φ_1, which is subsequently multiplied V^{res} to obtain the attention-activated feature map Z_c^r. The calculation formula is provided below

$$Z_c^r = \left(\frac{1}{1+\text{e}^{-\varphi_1}}\right) \odot V^{\text{res}} \tag{14}$$

where, Z_c^r is the feature map after cross-modal channel cross-self-attention activation, and $\odot$ is a Hadamard product operator.

For the prior data path, we use the same method to calculate the feature Z_c^p after cross-modal channel cross-self-attention activation.

Then the features Z_c^r and Z_c^p are activated by cross-modal space cross self-attention. We first utilize 1×1 convolutions to extract features from input features with size $Z_c^r \in \mathbb{R}^{C\times H\times W}$, resulting in a feature vector referred to as the interaction vector V_s^{inter}. Global Average Pooling is applied to the interactive vector V_s^{inter} across the spatial dimension to generate the feature vector F_x, facilitating the capture of global feature information for each feature layer. The dimension F_x is altered using the projection function $\xi(\cdot) : \mathbb{R}^{C\times 1\times 1} \rightarrow \mathbb{R}^{1\times C}$. Subsequently, the vector G_s^r is obtained by employing the Softmax activation function, thereby enhancing the dynamic range of feature activation. The calculation formula is provided below

$$G_s^r = \text{Soft}\max(\xi(F_x)). \tag{15}$$

Then, the projective function $\mu(\cdot)$ is used to change the dimension of the interactive vector V_s^{inter}, to get the vector $Q_s^r = \mu(V_s^{\text{inter}})$.

The vectors Q_s^p and G_s^p of the prior path are obtained using the same method. Similarity calculations are then performed between features Q_s^r and G_s^p, as well as Q_s^p and G_s^r, to obtain similarity matrices. These similarity matrices are converted into weights and assigned to V^{res} and I^{res}, respectively, resulting in cross-modal spatially-aware self-attention activated features Z_s^r and Z_s^P. Finally, a comprehensive cross-modal fusion feature O^{VI} is obtained using the formula below

$$O^{\text{VI}} = \text{Concat}\big(\big(Z_s^r \oplus V^{\text{res}}\big)\big(Z_s^p \oplus I^{tes}\big)\big). \tag{16}$$

D. Boundary Optimization Algorithm

Compared with learning edge features in the feature map, the segmentation map contains richer edge feature information, which can be directly refined in the segmentation prediction map. Therefore, we add independent edge loss calculations to the model. As shown in Fig. 2(a), the boundary optimization module consists of a standard cross-entropy loss function l_{seg} [42] and an edge optimization loss function l_{edge} [35]

$$l_{\text{BOA}} = l_{\text{seg}}(g, y) + l_{\text{edge}}\big(g_{\text{edge}}, y_{\text{edge}}\big) \tag{17}$$

where g represents the ground truth, and y represents the prediction results. The boundary labels g_{edge} are generated from the semantic segmentation ground truth using the Laplacian operation. Similarly, a simple edge extractor is added to the top of the segmentation network using the Laplace operator, which can directly extract gradient information y_{edge}. To reduce the computational burden, the gradient information map is transformed into a single-edge probability map.

During training, we obtain the prognostic maps at scales 64×64, 128×128, and 256×256 from d_3, d_4 and d_5, which are denoted as y_3, y_4, and y_5, respectively. In addition, we denote the ground truths of the corresponding sizes as g_3, g_4 and g_5. During testing, we only use the output of the final layer y_5 as the model's prediction results. The total boundary loss is

$$\begin{aligned} l_{\text{BOA}} = {} & l_{\text{seg}}(g_3, y_3) + l_{\text{seg}}(g_4, y_4) + l_{\text{seg}}(g_5, y_5) \\ & + l_{\text{edge}}\big(g_{3\text{edge}}, y_{3\text{edge}}\big) + l_{\text{edge}}\big(g_{4\text{edge}}, y_{4\text{edge}}\big) \\ & + l_{\text{edge}}\big(g_{5\text{edge}}, y_{5\text{edge}}\big). \end{aligned} \tag{18}$$

The overall loss function of MCKTNet is

$$l = l_{CGF} + l_{\text{CSF}} + l_{\text{BOA}}. \tag{19}$$

IV. Comparison and Discussion

In this section, we conducted semantic segmentation experiments on remote sensing images for water bodies, vegetation, and buildings, respectively, by utilizing different combinations of data sources. Then, we conducted ablation studies on each module of the proposed method. Finally, through comparisons with other methods, we further validated the effectiveness and generalization ability of the proposed approach.

A. Dataset

ISPRS Potsdam Dataset and Vaihingen Dataset: The Potsdam dataset comprises 38 high-resolution aerial remote sensing images and their corresponding DSMs. From these, a random selection of 24 images is designated as the training set, with the remaining images serving as the test set. Both the images and DSMs have a spatial resolution of 5 cm and dimensions of 6000×6000 pixels. The Vaihingen dataset contains 33 high-resolution aerial remote-sensing images and their corresponding DSMs. These high-resolution images include

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the red (R), green (G), blue (B), and NIR bands. The original labeled data encompasses six major land cover classes. In this study, buildings are defined as the foreground, while all other objects are classified as the background. *GID Dataset:* The Gaofen image dataset (GID) is an open-source satellite remote sensing dataset. It comprises ten finely labeled land cover images and 150 coarsely labeled land cover images. All images are captured by the GF-2 satellite, with a pixel resolution of 4 m. Each image contains the four bands of R, G, B, and NIR. We obtained NDWI and NDVI data corresponding to IRRGB images by calculating between bands. We selected 150 large-scale images with a size of 7200×6800 to validate the proposed method and selected 80 multispectral images (IRRGB) and the corresponding normalized index data (NDVI, NDWI) as the training set, the remaining 70 for testing. In the experiments on water body detection, we redefined the original dataset by categorizing rivers, lakes, and ponds as water bodies, while assigning other categories as background. Subsequently, we combined this redefined dataset with NDWI to create the multimodal GID (Water Body) dataset. Similarly, in the experiments on vegetation detection, forest, grassland, and dry fields are divided into vegetation categories, while other categories are divided into background, and combined with NDVI to make the multimodal GID (Vegetation) dataset.

B. Implementation Details

The PyTorch library was used to implement and train MCKTNet, which was trained using Adam optimization, and we set the learning rate of our training phase to $1e^{-5}$. The Mobilenetv2 [30] backbone network was initialized with pre-trained weights from the ImageNet dataset, and the other parts were initialized randomly.

C. Evaluation Metrics

The evaluation of experimental outcomes was performed using metrics including overall accuracy (OA), $F1$-Score ($F1$), and mean intersection over union (mIoU). The calculation formulas for these metrics are provided below

$$OA = \frac{\sum_{k=1}^{N} \mathrm{TP}_k}{\sum_{k=1}^{N} \mathrm{TP}_k + \mathrm{FP}_k + \mathrm{TN}_k + \mathrm{FN}_k} \tag{20}$$

$$\mathrm{MIoU} = \frac{1}{N} \sum_{k=1}^{N} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k} \tag{21}$$

$$\mathrm{precision}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k} \tag{22}$$

$$\mathrm{Recall}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FN}_k} \tag{23}$$

$$F1\mathrm{Score} = \frac{2 \times \left(\mathrm{precision}_k \times \mathrm{Recall}_k\right)}{\mathrm{precision}_k + \mathrm{Recall}_k} \tag{24}$$

where TP_k, FP_k, FN_k, and TN_k indicate the true positive, false positive, true negative, and false negatives, respectively, for object indexed as class k.

D. Ablation Study

Ablation studies were conducted to assess the effectiveness of key components in the proposed MCKTNet. Tables I–VI provide a detailed account of these experiments and their results. The symbol indicates that the corresponding module was retained, while the absence of this symbol indicates that the module was removed. All other settings, including loss functions and optimizers, remained consistent with the complete MCKTNet.

1) The effectiveness of CGSF: We removed the CGSF module and directly fed the features r and t extracted from different modality branches into the multimodal cross-self-attention module (CAM) for cross-modal feature fusion. As observed in Table I, the inclusion of CGSF significantly improves segmentation accuracy across multiple datasets. To demonstrate the role of CGSF more intuitively, we present the CAM output feature heatmaps from the model trained on the Potsdam (Building) dataset. As shown in Fig. 4, Conv Layer 1 through Conv Layer 5 correspond to heatmaps of the CAM output features O_1^{VI} through O_5^{VI}, respectively. The color of the heatmap closer to red indicates a higher feature response of the network at that point. Each pair of Conv Layer includes two images: the left image shows the result removed CGSF, while the right image shows the result with CGSF added. From the analysis of the figures, it can be observed that the shallow features extracted by the network (Conv Layer 1 and Conv Layer 2) primarily focus on geometric characteristics such as object shape and edge contours. In contrast, the deep features (Conv Layer 4 and Conv Layer 5) concentrate on the semantic attributes of the objects. Therefore, we incorporated the CGF module in the shallow layers for the transfer of geometric features, while the CSF module was included in the deeper layers for semantic feature transfer. In addition, both the CGF and CSF modules were implemented in the intermediate layer (Conv Layer 3). As shown in Fig. 4, the addition of CGSF significantly enhances the model's ability to extract shallow geometric features and deep semantic features. As shown in Fig. 5, for the water body detection task, incorporating the CGSF module enhances the model's ability to recognize complex texture regions and easily confused water-land boundaries. The experiments demonstrate that reducing the data distribution discrepancy between different modal features before cross-modal feature fusion can enhance the effectiveness of the fusion process.

CGSF Without CGF: To validate the effectiveness of CGF, we replaced the CGF module in the shallow layers of the network with the CSF module, resulting in a unified CSF module for cross-modal data distribution learning throughout the network. As indicated in Table I, the removal of the CGF module from the model leads to a decrease in segmentation accuracy, confirming that geometric feature transfer is more suitable in the shallow layers of the network.

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TABLE I
ABLATION EXPERIMENT RESULTS CONSIDERING DIFFERENT MODULES OF MCKTNet. THE BOLD VALUES INDICATE THE BEST RESULTS

CGSF			Postdam (Buiding)		GID (Water body)		GID (Vegetation)	
CGF	CSF	AF	OA(%)	mIoU(%)	OA(%)	mIoU(%)	OA(%)	mIoU(%)
			96.82	92.11	95.43	87.04	90.25	81.28
	√	√	97.39	93.54	96.05	88.62	91.12	82.19
√		√	97.52	93.71	96.32	88.93	91.46	82.34
√	√		97.07	93.25	95.91	88.17	90.66	81.78
√	√	√	**97.73**	**93.78**	**96.82**	**89.41**	**92.21**	**83.52**

TABLE II
ABLATION EXPERIMENT RESULTS OF DIFFERENT CHARACTERISTIC DISTRIBUTION LOSS. THE BOLD VALUES INDICATE THE BEST RESULTS

CGSF		GID (Water body)		
CGF	CSF	F1(%)	OA(%)	mIoU(%)
MSE	MSE	94.41	96.51	89.15
KL	MSE	93.89	96.12	88.92
KL	KL	94.32	96.38	89.03
MSE	KL	**94.70**	**96.82**	**89.41**

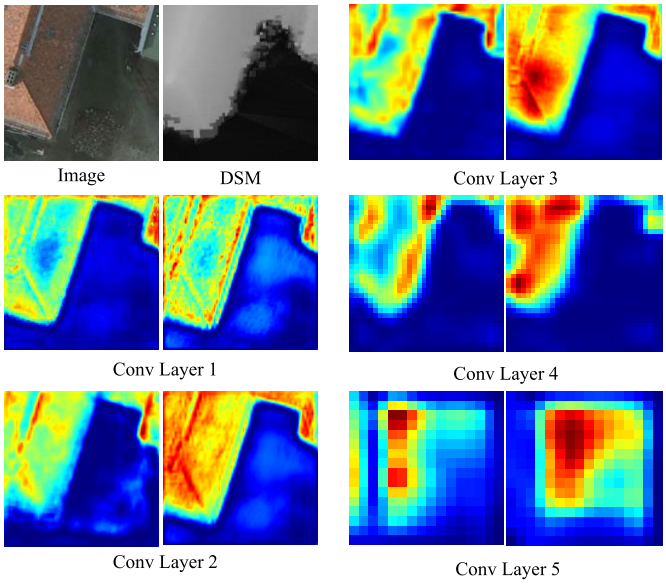


Fig. 4. Heatmaps of the CAM output features trained on the potsdam (building) dataset.

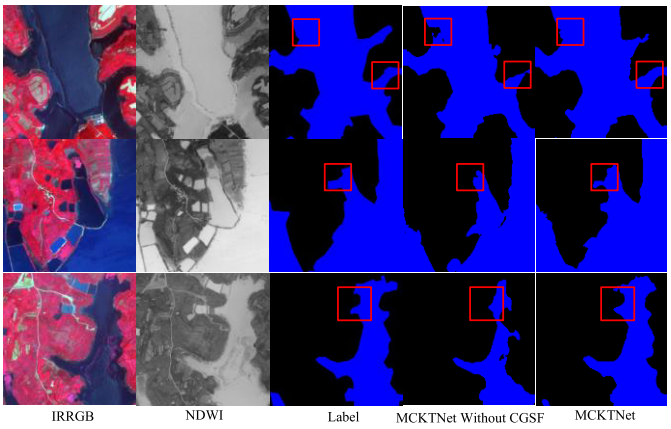


Fig. 5. Semantic segmentation comparison for MCKTNet and MCKTNet without CGSF on the GID (water body) dataset.

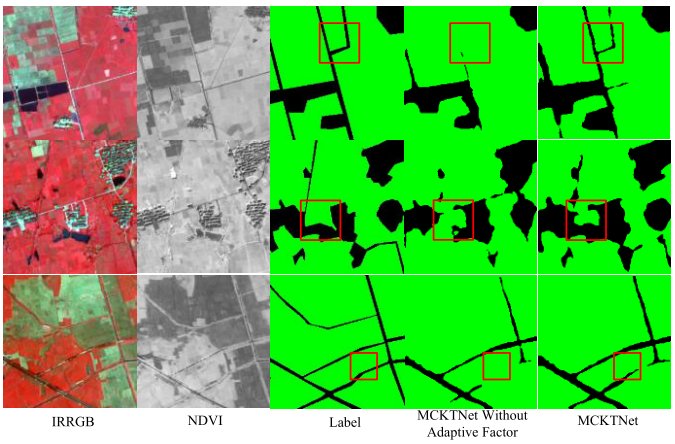


Fig. 6. Semantic segmentation comparison for MCKTNet and MCKTNet without AF on the GID (vegetation) dataset.

CGSF Without CSF: Similarly, we replaced the CSF module in the deep layers of the network with the CGF module, resulting in a unified CGF module for cross-modal data distribution learning throughout the network. As observed in Table I, the removal of the CSF module from the model leads to a decrease in segmentation accuracy, demonstrating that semantic feature transfer is more suitable in the deep layers of the network.

*CGSF Without AF:*To validate the effectiveness of incorporating an AF into the CGSF module, we tested the model's performance without the AF. As depicted in Fig. 6, the inclusion of the AF significantly enhances the model's ability to recognize detailed features. This demonstrates that during cross-modal feature transfer, the addition of an AF can effectively mitigate the issues of knowledge redundancy and the loss of intrinsic features caused by large data distribution discrepancies between different data sources during mutual feature transfer. Acting as a controller, the AF enables gradual cross-modal knowledge transfer, which aids in preserving the detailed features of different data sources. As observed in Table I, the inclusion of the AF effectively enhances the model's semantic segmentation performance.

2) *Characteristic Distribution Loss:* Both mse loss and KL divergence loss can achieve cross-modal knowledge transfer. We derived the most suitable cross-modal

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TABLE III
ABLATION EXPERIMENT RESULTS CONSIDERING THE THREE MCKTNet MODULES. THE BOLD VALUES INDICATE THE BEST RESULTS

CGSF	CAM	BOA	Postdam (Buiding)		GID (Water body)		GID (Vegetation)	
			OA(%)	mIoU(%)	OA(%)	mIoU(%)	OA(%)	mIoU(%)
√	√		97.34	93.61	96.59	89.23	91.80	82.97
√		√	94.16	90.83	92.35	84.67	87.39	77.26
	√	√	96.82	92.11	95.43	87.04	90.25	81.28
√	√	√	**97.73**	**93.78**	**96.82**	**89.41**	**92.21**	**83.52**

TABLE IV
COMPARISON OF mIoU FOR DIFFERENT COMBINATIONS OF CHANNEL AND SPATIAL ATTENTION SUBMODULES (%). THE BOLD VALUES INDICATE THE BEST RESULTS

CAM	Postdam (Buiding)	GID (Water body)	GID (Vegetation)
channel + spatial	**93.78**	**89.41**	**83.52**
spatial + channel	93.54	88.76	82.69
channel & spatial in parallel	93.19	88.04	82.47

TABLE V
COMPARISON OF mIoU FOR EDGE OPTIMIZATION APPLIED TO FEATURE MAPS OF DIFFERENT SCALES (%). THE BOLD VALUES INDICATE THE BEST RESULTS

BOA	Postdam (Buiding)	GID (Water body)	GID (Vegetation)
BOA_5	93.27	88.24	82.81
BOA_4,5	93.69	89.30	83.17
BOA_3,4,5	**93.78**	**89.41**	**83.52**
BOA_2,3,4,5	93.72	89.36	83.29

feature distribution loss calculation method for each module by using different loss functions on the CGF module and CSF module. As shown in Table II, taking water body detection as an example, mse loss in CGF can increase mIoU by 0.38% compared with KL loss. Compared with mse loss, KL loss on CSF can increase mIoU by 0.26%. The difference in feature transfer capabilities of mse loss and KL divergence loss in cross-modal knowledge transfer stems from the fact that mse loss, by minimizing the squared error between predicted and true values, focuses more on numerical accuracy and local details, making it more suitable for transferring shallow features such as geometry. In contrast, KL divergence loss, by measuring the distance between two probability distributions, emphasizes the overall similarity and structure of the distributions, capturing deep semantic information in the data and thus more effectively transferring deep features such as semantics.

3) *The Effectiveness of CAM:* To validate the effectiveness of the CAM module, we tested a model that does not utilize CAM but instead directly concatenates the features extracted from different modality branches for feature fusion. To more intuitively demonstrate the effectiveness of different modules, as shown in Fig. 7, we visualize the feature heatmaps generated by the final output layer d_5 of the model in the building detection experiment. Fig. 7(d) represents the heatmap without the CAM, while Fig. 7(f) presents the heatmap of the MCKTNet with CAM included. The results indicate that the model with the added CAM module exhibits a stronger and more balanced response to building targets with significant morphological variations. As shown in Fig. 8, in the building segmentation experiment, the inclusion of CAM improved the model's ability to recognize easily confused objects, particularly mitigating the occlusion issue caused by trees and enhancing the model's robustness against interferences. As indicated in Table III, not using CAM resulted in a significant drop in semantic segmentation accuracy, suggesting that cross-modal feature transfer alone is insufficient for effective multimodal feature fusion. While the CGSF module reduces differences in data distributions to enable multimodal data fusion, it fails to effectively identify complementary features across different modalities. In addition, Table III illustrates that cross-modal feature transfer serves as an effective preprocessing step before multimodal feature fusion, as the difficulty of cross-modal data fusion decreases with reduced distributional differences between the modalities.

The CAM module establishes correlations between different modalities from both channel and spatial perspectives. As each module has different functions, the order of the channel and spatial attention submodules can influence overall performance. We compare three different ways of arranging the channel and spatial attention submodules: sequential channel-spatial, sequential spatial-channel, and parallel use of both attention modules. Table IV summarizes the mIoU values for each arrangement across three datasets. The results indicate that the sequential channel-spatial configuration outperforms the others slightly. From the perspective of spatial features, channel attention emphasizes global semantic features, while spatial attention focuses on local details. Therefore, applying channel attention first to select significant global semantic features, followed by spatial attention to concentrate on localized regions at a finer level, aligns well with the human visual system's progressive focus on finer details. This stepwise refinement effectively enhances the representational capacity of the features.

4) *The Effectiveness of BOA:* To validate the effectiveness of the BOA module, we tested a model that solely relies on the cross-entropy loss function l_{seg}

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TABLE VI
COMPARISON OF ABLATION EXPERIMENTS USING INPUT DATA FROM DIFFERENT MODALITIES. THE BOLD VALUES INDICATE THE BEST RESULTS

Input data	Postdam (Buiding)		GID (Water body)		GID (Vegetation)	
	OA(%)	mIoU(%)	OA(%)	mIoU(%)	OA(%)	mIoU(%)
RGB	97.32	91.85	88.76	80.62	87.52	77.46
IRRGB	97.84	92.67	94.69	86.16	88.32	80.93
NDWI	-	-	93.43	84.15	-	-
NDVI	-	-	-	-	67.24	50.71
IRRGB+ DSM	**97.73**	**93.78**	-	-	-	-
IRRGB+ NDWI	**-**	**-**	**96.82**	**89.41**	-	-
IRRGB+ NDVI	**-**	**-**	**-**	**-**	**92.21**	**83.52**

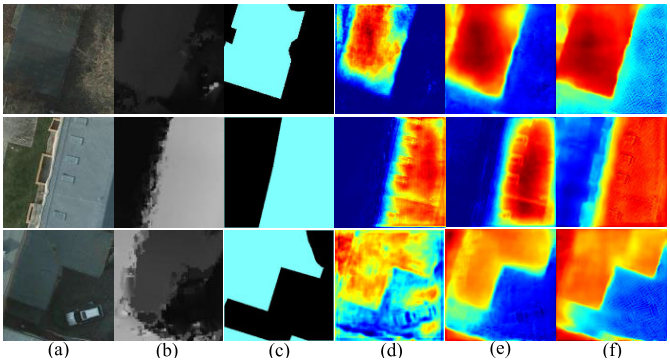


Fig. 7. Visualization of heatmaps for models trained on the potsdam (building) dataset. (a) Represents the input image, (b) represents the input DSM, (c) represents the label, (d) represents the model without CAM, (e) represents the model without BOA, and (f) represents the MCKTNet that includes both CAM and BOA.

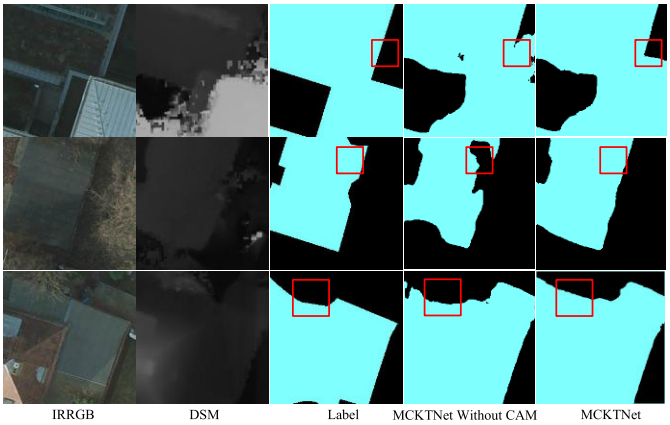


Fig. 8. Semantic segmentation comparison for MCKTNet and MCKTNet without CAM on the postdam (buiding) dataset.

for calculating the loss of model predictions, without employing the edge optimization loss function l_{edge}. Fig. 7(e) illustrates the heatmaps when the BOA module is removed, whereas Fig. 7(f) presents the heatmaps of the MCKTNet with BOA included. The results indicate that the heatmaps incorporating the BOA module exhibit a stronger response to the edge regions of the target. Table III demonstrates after adding the BOA module to the baseline network, the performance of OA on the Potsdam (Buiding) is improved by 0.39%. The BOA module employs multiscale loss calculation. We designed an ablation experiment to demonstrate the rationality and superiority of this approach, obtaining the results listed in Table V, where BOA_N indicates that dN was used (Such as BOA_5 indicates that only d5 was used, while BOA_4, 5 indicates that d4 and d5 were used). The results show the importance of using d3–d5 in BOA. As deeper layers, such as d1 and d2, have very small feature scales, they contain almost no boundary information and therefore do not require additional supervision.

5) In addition, to verify the influence of different modal data on the segmentation accuracy, we conduct extensive ablation experiments on input data of different modalities. As shown in Table VI, we used RGB and IRRGB images as the single data source, respectively, for model training and testing on Potsdam (Buiding), GID (Water body), and GID (Vegetation) datasets. RGB images contain rich color and texture features, but they are easily affected by environmental factors and other factors, and the ability to recognize confusing objects is insufficient. IRRGB data increases the NIR band compared with RGB data, which improves the recognition ability of the model for different ground objects. The MIoU values of the three datasets are increased by 0.82%, 5.54%, and 3.47%, respectively. NDWI and NDVI data can highlight the characteristics of water and vegetation, respectively. Threshold segmentation of NDWI and NDVI data is a common detection algorithm for water and vegetation. However, using a single NDWI and NDVI data cannot obtain the texture features and semantic information of the target, and it is highly dependent on the algorithm parameter Settings, and the robustness is insufficient. As shown in Fig. 9, the vegetation detection method based on threshold segmentation of NDVI data can identify smaller vegetation areas but is insufficient for detecting yellowed vegetation. The vegetation segmentation method based on RGB images cannot recognize spatial detail features, and the edge of the segmentation result is blurred. Our method improves the accuracy of semantic segmentation by combining different modal data and comprehensively utilizing their respective advantages.

E. Quantitative Comparison Diverse Methods

Our method was compared with state-of-the-art semantic segmentation methods and included classic multiscale context feature fusion methods such as FCN [36], U-Net [37],

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TABLE VII

EXPERIMENT RESULTS ON THE POSTDAM (BUIDING) DATASET, GID (WATER BODY) DATASET, AND GID (VEGETATION) DATASET. THE BOLD VALUES INDICATE THE BEST RESULTS, AND THE UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS

Methods	Postdam (Buiding)			GID (Water body)			GID (Vegetation)		
	F1(%)	OA(%)	mIoU(%)	F1(%)	OA(%)	mIoU(%)	F1(%)	OA(%)	mIoU(%)
FCN-8s [36]	88.49	91.75	86.33	82.12	83.72	75.13	82.15	84.11	74.72
U-Net [37]	90.68	93.58	87.15	82.24	83.85	75.42	82.39	84.30	74.93
DeepLabv3+ [38]	91.03	93.91	87.62	84.18	85.27	76.91	84.27	85.49	75.06
MANet [16]	92.77	95.13	89.46	86.34	87.40	79.13	85.73	86.20	76.15
UNetFormer [41]	93.02	95.98	90.73	87.69	88.86	80.90	86.12	87.46	77.28
GCDNet [42]	93.60	96.42	91.26	89.07	90.19	81.83	86.35	87.73	77.92
PACSCNet [39]	94.09	97.14	91.95	90.35	91.16	83.42	87.83	88.35	78.54
CMGFNet [40]	94.28	97.25	92.35	90.94	92.27	84.10	88.04	88.76	78.82
FTransUNet [43]	94.75	97.31	93.14	91.23	92.84	85.05	88.83	88.92	79.63
CMX [44]	94.06	97.12	92.29	91.67	93.05	85.29	89.17	89.08	80.23
CMFPNet [46]	94.92	97.46	93.35	92.54	93.98	86.75	89.10	89.02	80.18
MultiSenseSeg [47]	93.86	96.62	91.30	93.22	94.75	87.13	90.82	90.63	81.04
MCKTNet (Ours)	**95.38**	**97.73**	**93.78**	**94.70**	**96.82**	**89.41**	**92.53**	**92.21**	**83.52**

TABLE VIII

COMPUTATIONAL COMPLEXITY ANALYSIS FOR MEASURING 512 × 512 IMAGES ON A SINGLE NVIDIA GEFORCE RTX 3090TI GPU. THE mIoU VALUE IS THE RESULT ON THE ISPRS POTSDAM (BUIDING) DATASETS

Methods	Multimodal	Baseline	FLOPs (G) ↓	Parameter (M) ↓	Memory (MB) ↓	mIoU (%) ↑
Deeplabv3+	N	ResNet-101	138.63	40.35	4336	87.62
UNetFormer	N	ResNet-50	46.04	24.20	2483	90.73
CMTFNet	N	ResNet-50	32.85	30.07	3472	91.12
PACSCNet	Y	ResNet-50	276.54	359.64	9847	91.95
CMX	Y	SegFormer	136.62	141.46	7359	92.29
FTransUNet	Y	ResNet-50	45.21	160.88	3463	93.14
CMFPNet	Y	CMFPNet	21.28	22.82	3349	93.35
MCKTNet (Ours)	Y	Mobilenetv2	11.89	13.51	2239	93.78

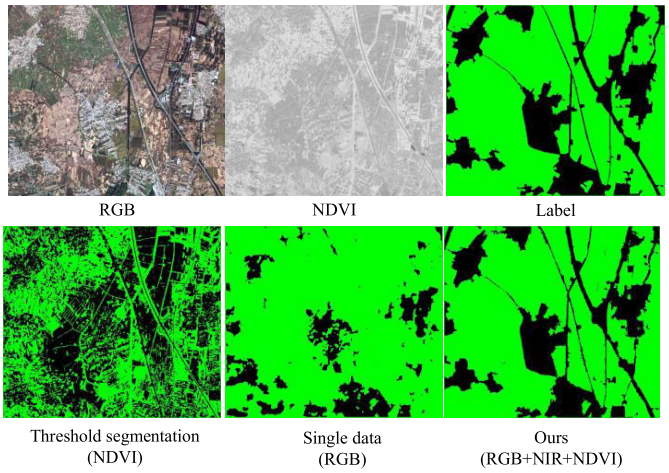


Fig. 9. Results of vegetation detection using different modal data.

and Deeplabv3+ [38]. As well as advanced remote sensing image segmentation methods that utilize transformers for global feature extraction, such as MANet [16], UNetFormer [41], CMTFNet [45], and global context dependencies aware network (GCDNet) [42]. These methods only consider RGB image information. These advanced single-modality methods can demonstrate the performance improvements brought by using multimodal prior data. In addition, we compared MCKTNet with state-of-the-art multimodal feature fusion networks, including PACSCNet [39], CMGFNet [40], FTransUNet [34], cross-modal fusion for RGB-X semantic segmentation (CMX) [44], CMFPNet [46], and MultiSenseSeg [47]. The test results are shown in Table VII.

UNetFormer uses a lightweight ResNet18 encoder and a transformer-based decoder to establish long-range dependencies, improving the utilization of both global and local information. CMGFNet employs end-to-end cross-modal gated fusion and multilevel feature fusion techniques, to improve the extraction of building footprints from VHR remote sensing images and DSM data. PACSCNet leverages a dual-pyramid symmetric cascade decoder and a multiscale feature extraction module to enhance segmentation accuracy by effectively harnessing multimodal contextual features. CMFPNet employed two independent feature extraction branches to extract the semantic information of offshore aquaculture areas from different remote sensing image modalities. CMFPNet designed LGPB to capture both local and global feature information to enhance the recognition ability of the model under complex and variable background interference. To address the large number of parameters and expensive computational costs of multimodal semantic segmentation models, MultiSenseSeg employs multiple lightweight modality-specific experts (MSEs), an adaptive multimodal matching (AMM) module,

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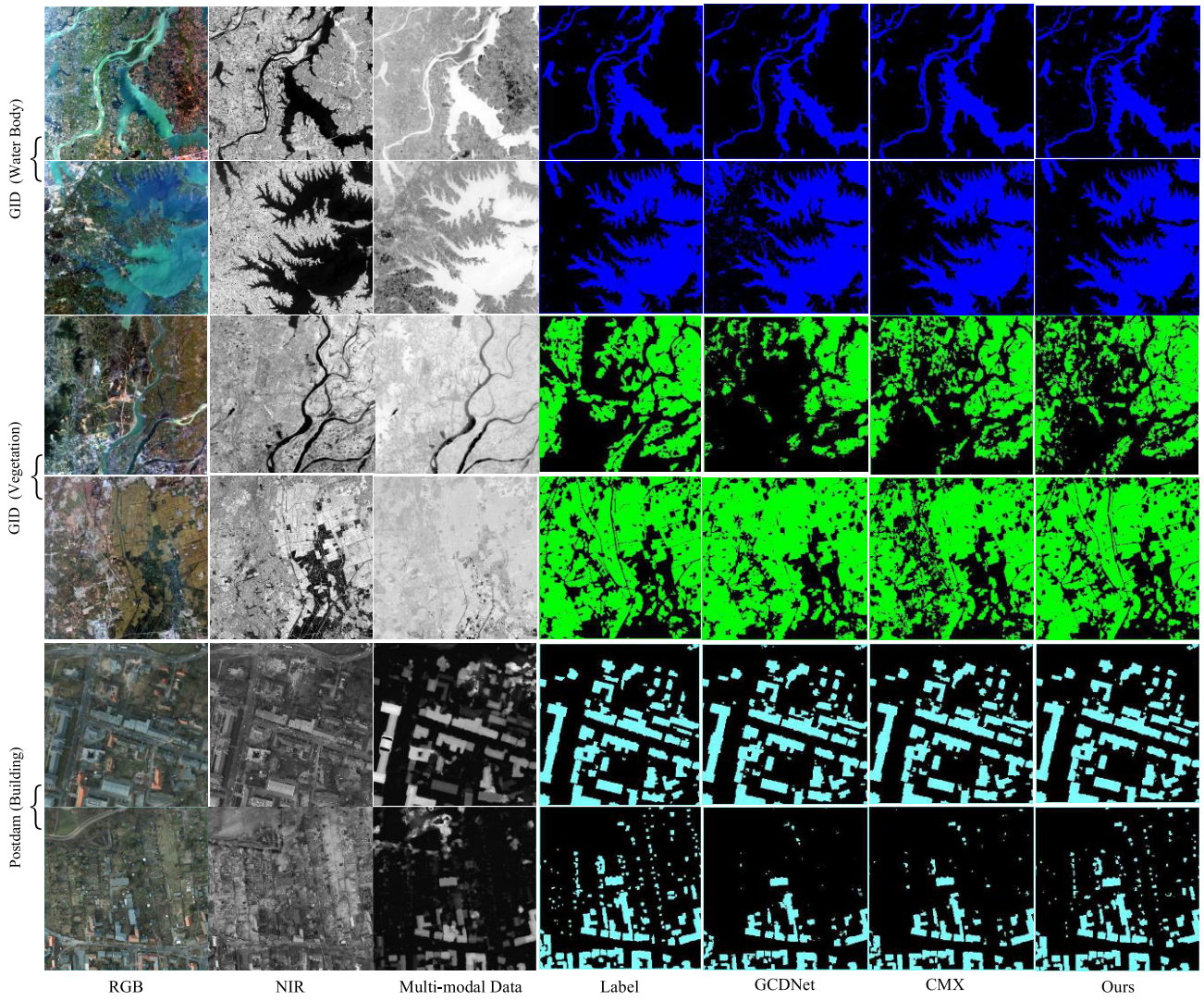


Fig. 10. Complete results of different methods. (First two rows) are the test results of the GID (water body) dataset, the multimodal data used is NDWI. (Middle two rows) are the test results of the GID (vegetation) dataset, the multimodal data used is NDVI. (Last two rows) are the test results of the postdam (building) dataset, the multimodal data used is DSM.

and a single feature extraction pipeline to efficiently model intramodal and intermodal relationships.

As shown in Table VII, the numeric scores for the ISPRS Potsdam (Buiding), GID (Water body), and GID (Vegetation) datasets demonstrated that MCKTNet delivers high accuracy, exceeding other networks in the $F1$, OA, and mIoU by a significant margin. The experimental results of different methods on the Potsdam (Buiding) dataset demonstrated that MCKTNet delivers the highest $F1$ of 95.38%, OA of 97.73%, and mIoU of 93.78%. The experimental results of different methods on the GID (Water body) dataset demonstrated that MCKTNet delivers the highest $F1$ of 94.70%, OA of 96.82%, and mIoU of 89.41%. The experimental results of different methods on the GID (Vegetation) dataset demonstrated that MCKTNet delivers the highest $F1$ of 92.53%, OA of 92.21%, and mIoU of 83.52%. In terms of test metrics, MCKTNet achieves the highest segmentation accuracy in different semantic segmentation tasks, indicating that our method not only can extract effective complementary information from multimodal features but also has strong generalization capabilities.

The comparison of experimental results between MCKTNet and the state-of-the-art single-modal remote sensing image semantic segmentation network (GCDNet) and the state-of-the-art multimodal remote sensing image semantic segmentation network (CMX) is shown in Fig. 10. In terms of segmentation performance, MCKTNet has a strong recognition capability for confusing objects, such as paddy fields and dry fields, rivers with different colors. It is effective at segmenting small objects such as tiny buildings hidden in forests. In addition, it achieves excellent segmentation results for complex edge contours of objects.

Experimental results demonstrate that MCKTNet outperforms other models in terms of its ability to effectively extract intricate image features, accurately discern target locations and boundaries, and exhibit superior generalization capabilities. It successfully mitigates the challenges posed by noise interference and the loss of detailed features of different

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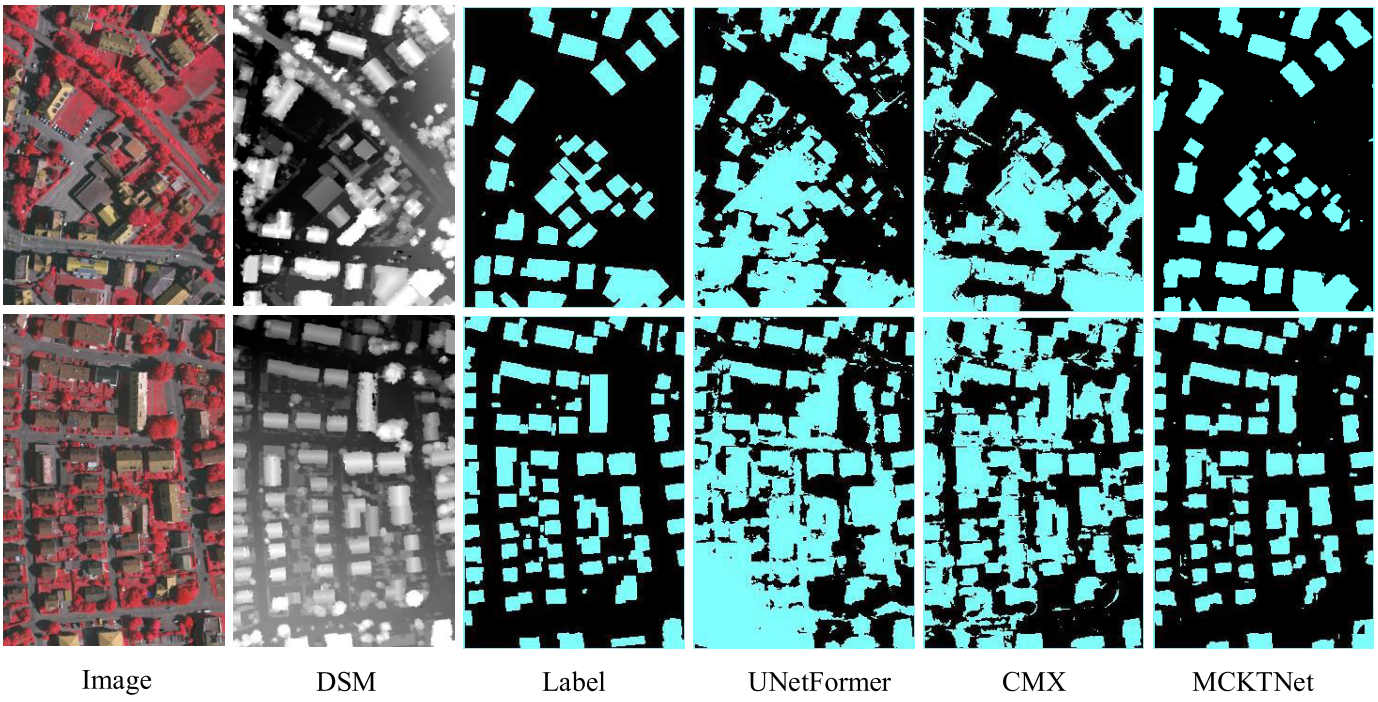


Fig. 11. Results of zero-shot building detection on Vaihingen (building) dataset using the model trained on postdam (building) dataset.

modal data, which often leads to inaccurate target localization and boundary pixel segmentation. Moreover, it addresses the issue of inadequate model robustness in semantic segmentation tasks involving multimodal remote sensing images. Our approach exhibits excellent performance on diverse datasets and achieves superior segmentation outcomes compared to alternative networks.

F. Model Parameters and Computation Complexity Analysis

We evaluate the computational complexity of the proposed MCKTNet using the following evaluation metrics: the floating point operation count (FLOP), the number of model parameters, and the memory footprint. We fed input data of size 512×512 into all models and also evaluated the computational complexity and parameter sizes of various methods in the same runtime environment, and the results are shown in Table VIII. From Table VIII it can be seen that the computational complexity of multimodal models is generally higher than that of single-modal models. MCKTNet has 11.89G FLOPs, 13.51 M Parameters, and 2239 MB Memory, which is far less than the model complexity of other multimodal fusion methods, and achieves the optimal segmentation accuracy, showing the superior performance of MCKTNet.

G. Generalization Ability

Due to the rich textural and semantic features of multispectral images, object recognition algorithms based on them are susceptible to variations in different scenes and environmental factors, often lacking generalization ability across datasets. Multimodal prior data typically emphasize the basic physical attributes of objects and are less affected by different scenes and environmental factors. Therefore, if a model can effectively learn features from prior data, it can enhance its cross-dataset generalization capability.

TABLE IX
QUANTITATIVE COMPARISON OF ZERO-SAMPLE BUILDING DETECTION RESULTS ON VAIHINGEN (BUILDING) DATASET USING THE MODEL TRAINED ON POSTDAM (BUILDING) DATASET

Methods	Multimodal	Vaihingen (Buiding)		
		F1(%)	OA(%)	mIoU(%)
UNetFormer	N	65.73	75.26	58.49
GCDNet	N	67.29	76.90	60.21
PACSCNet	Y	67.84	77.36	61.75
CMX	Y	69.92	79.64	63.54
MultiSenseSeg	Y	71.58	81.22	65.39
MCKTNet Without CGSF	Y	70.40	80.68	64.75
MCKTNet	Y	89.31	94.80	87.02

To verify the generalization ability of MCKTNet across different datasets, we trained MCKTNet on the Potsdam (Building) dataset and then conducted a zero-shot building detection experiment on the Vaihingen (Building) dataset using the obtained model without any debugging or retraining. As shown in Fig. 11, MCKTNet is capable of effectively detecting buildings on the Vaihingen (Building) dataset under zero-shot conditions. Table IX demonstrates that the CGSF module can significantly improve the model's cross-modal feature learning ability. Compared with other state-of-the-art semantic segmentation algorithms, MCKTNet exhibits the strongest domain generalization capability.

V. CONCLUSION

This article proposes a multiscale cross-modal knowledge transfer network for semantic segmentation of remote sensing images. First, a cross-modal knowledge transfer network is

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designed to transfer semantic and geometric features between different modalities, aiming to reduce the difference in feature distribution between these modalities. Subsequently, a cross-attention mechanism is used to achieve a complementary fusion of cross-modal features. Finally, to obtain more accurate object boundaries, we design a multiscale edge-aware optimization module that can directly refine edge details based on pixel-level predictions. We use transfer learning to significantly reduce the redundancy between different modal data and achieve bidirectional transfer of semantic and geometric information. Our method focuses on enhancing the representation of backbone features rather than network prediction and is therefore more straightforward. Furthermore, our method enables simultaneous bidirectional training without requiring pre-training of the network or components. Extensive experiments demonstrate the good performance, robustness, and generalization capabilities of MCKTNet. It achieves the best segmentation accuracy in multiple remote sensing image semantic segmentation tasks, while the computational complexity is significantly lower than other multimodal fusion models.

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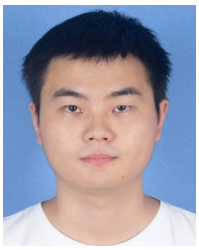
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